Julio Majesty Rasjid

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RESEARCH INTERESTS

Bayesian inference, distribution theory, parametric statistical inference, statistical modelling

EDUCATION

Master of Science in Statistics Oct 2024 – Mar 2026
Universiti Malaya Kuala Lumpur, Malaysia

- GPA: 3.97/4.00 (With Distinction)
- Thesis: *On the Discretisation of the Odd Power Generalised Weibull-G Family of Distributions*
- Relevant Coursework: Bayesian Statistics, Probability Theory, Statistical Inference, Generalized Linear Models, Multivariate Analysis

Bachelor of Science in Mathematics Sep 2018 – Jan 2022
Universitas Indonesia Depok, Indonesia

- GPA: 3.91/4.00 (Cum Laude), Best Graduate
- Thesis: *Alpha Power Inverse Weibull Distribution*
- Relevant Coursework: Mathematical Statistics, Introduction to Probability Theory, Survival Analysis, Regression Analysis, Real Analysis

RESEARCH EXPERIENCE

Master's Thesis Mar 2025 – Jan 2026
Institute of Mathematical Sciences, Universiti Malaya Kuala Lumpur, Malaysia

Developed a new family of discrete lifetime distributions by discretising the odd power generalised Weibull-G (OPGW-G) continuous family, motivated by the limited availability of flexible discrete models for count-type failure and reliability data. Derived closed-form probability mass functions and key statistical properties. Constructed maximum likelihood estimation frameworks and conducted Monte Carlo simulation studies across varying sample sizes to evaluate finite-sample behaviour and consistency of estimators. Proposed models demonstrated competitive or superior fit over established discrete competitors on multiple real-world data sets, assessed via information criteria (AIC and BIC) and goodness-of-fit tests.

Undergraduate Thesis Sep 2021 – Dec 2021
Department of Mathematics, Universitas Indonesia Depok, Indonesia

Conducted a systematic analytical study of the alpha power inverse Weibull distribution, deriving and examining key statistical properties including the survival function, hazard rate, moments, and quantile function. Established maximum likelihood estimation procedures. Applied the model to gastric cancer survival data, benchmarking its fit against a competing lifetime model. Results were published in ITM Web of Conferences and recognised with the Best Presenter Award at ISMART 2022.

PUBLICATIONS

Conference Proceedings

- **Rasjid, J. M.**, Nurrohmah, S., & Fithriani, I. (2024). Alpha power inverse Weibull distribution: A new lifetime distribution with application to gastric cancer data. *ITM Web of Conferences, 61*, 01009. [Link]

Manuscripts in Preparation

- **Rasjid, J. M.**, & Ng, C. M. *On the Discretisation of the Odd Power Generalised Weibull-G Family of Distributions.* Manuscript in preparation.

CONFERENCE PRESENTATIONS

- **Rasjid, J. M.**, & Ng, C. M. (2025). *On the discretisation of the odd power generalised Weibull-G family of distributions* [Oral presentation]. *3rd International Conference on Data Science (ICDS)*, University of Colombo, Colombo, Sri Lanka (Hybrid).

- **Rasjid, J. M.**, Nurrohmah, S., & Fithriani, I. (2022). *Alpha power inverse Weibull distribution* [Oral presentation]. *7th International Science and Mathematics Academics Research Talks (ISMART)*, Universitas Indonesia, Depok, Indonesia.

TEACHING EXPERIENCE

Graduate Teaching Assistant Apr 2025 – Jun 2025
Institute of Mathematical Sciences, Universiti Malaya Kuala Lumpur, Malaysia

- Delivered tutorial sessions for *Probability and Statistics I* for undergraduate students.
- Guided students through structured problem-solving exercises.

Mathematics Teacher Jul 2022 – Present
Alta Global School Jakarta, Indonesia (Remote)

- Deliver instruction in Cambridge AS & A Level Mathematics (Pure Mathematics; Probability & Statistics) and SAT Mathematics for secondary school students with emphasis on rigorous mathematical reasoning and conceptual depth.
- Develop instructional materials and problem sets to strengthen conceptual understanding.

Teaching Assistant Sep 2019 – Dec 2021
Department of Mathematics, Universitas Indonesia Depok, Indonesia

Courses: Calculus I & II, Linear Algebra, Mathematical Statistics I, Numerical Methods, Logic and Set Theory, Real Analysis II.

- Delivered tutorial sessions for undergraduate mathematics courses with class sizes of approximately 40 students.
- Developed tutorial problem sets to strengthen students' understanding.
- Graded coursework and provided feedback to students.

AWARDS & HONOURS

- **Best Graduate**, Undergraduate Mathematics Programme, Faculty of Mathematics and Natural Sciences, Universitas Indonesia (2022)

- **Best Presenter**, 7th International Science and Mathematics Academics Research Talks, Faculty of Mathematics and Natural Sciences, Universitas Indonesia (2022)

- **3rd Place**, National Mathematics Paper Competition, Mathematics Student Association, Universitas Gadjah Mada (2021)

TECHNICAL SKILLS

Statistical Computing: R (simulation studies, likelihood-based inference, goodness-of-fit testing, data analysis, MCMC)

Scientific Writing: LaTeX (mathematical typesetting)

LANGUAGES

Indonesian (Native), English (IELTS 7.0), Chinese (Elementary, HSK 3 equivalent)